

Exploring spatio-temporal features of Foundation Models for Aerial Depth and Semantic Scene Completion

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Abstract

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037

001 Aerial 3D perception must recover metric geometry and
002 semantic occupancy despite altitude-dependent scale,
003 oblique viewpoints, and limited onboard sensing. We
004 study what frozen image and video foundation rep-
005 resentations transfer to monocular depth estimation
006 (MDE) and semantic scene completion (SSC) on Occu-
007 Fly. Under a common 384×384 protocol, we progress
008 from last-layer linear probes to four-level DPT decod-
009 ing, altitude/intrinsics-conditioned FiLM-DPT, dense
010 temporal aggregation, and pose-aware multiview lifting.
011 We additionally introduce matched frozen DINOv3 con-
012 trols to separate conclusions about V-JEPA 2.1 from
013 properties shared by large self-supervised ViTs. Cor-
014 recting an anomalous historical DPT baseline changes
015 the central interpretation: multi-level decoding im-
016 proves over linear probing, while FiLM must be as-
017 sessed against a healthy, capacity-matched DPT rather
018 than against a failed run. The matched DINOv3 FiLM-
019 DPT reaches 4.358 m test RMSE, compared with
020 4.770 m for V-JEPA 2.1 FiLM-DPT, 6.730 m for the
021 corrected V-JEPA plain DPT, and 7.795 m for its lin-
022 ear probe. SSC experiments show that geometry, lift-
023 ing, and semantic context must remain aligned: across
024 three seeds, combining DINOv3 MVFormer++ geom-
025 etry with full-resolution VoxNT and long-tail optimiza-
026 tion reaches 49.33 ± 1.17 SC IoU and 5.22 ± 0.35 union-
027 present SSC mIoU. A UAV-oriented 11-class remap-
028 ping reaches 9.32 validation mIoU, indicating that
029 sparse, visually overlapping fine classes explain part—
030 but not all—of the low semantic score. Temporal aggre-
031 gation is inconsistent on snapshot neighborhoods; its
032 value may require continuous video with explicit cor-
033 respondence. Our study therefore characterizes where
034 frozen representations expose aerial geometry, where
035 decoder structure and camera metadata matter, and
036 which apparent gains do not survive controlled com-

1. Introduction

038

039 Aerial 3D scene understanding is central to au-
040 tonomous uncrewed aerial vehicle (UAV) operation
041 in cargo delivery, search and rescue, inspection, and
042 human-UAV collaboration. In these settings, a percep-
043 tion system must infer not only visible image semantics,
044 but also metric depth, free space, and the semantic oc-
045 cupancy of partially observed volumes. RGB cameras
046 are attractive for UAVs because they are lightweight,
047 inexpensive, and compatible with strict size, weight,
048 and power constraints. However, monocular 3D per-
049 ception remains difficult: image formation collapses
050 scale, depth, and occlusion into a two-dimensional
051 signal, and aerial views introduce altitude-dependent
052 scale changes, oblique/top-down viewpoints, weak tex-
053 ture, and occlusion patterns that differ from ground-
054 level driving scenes.

055 Most progress in semantic scene completion (SSC)
056 and camera-based 3D perception has been driven by
057 indoor and autonomous-driving benchmarks. SSC-
058 Net introduced semantic completion from depth input
059 [33]; MonoScene extended the task to monocular RGB
060 [8]; and subsequent methods such as VoxFormer [21],
061 TPVFormer [17], SurroundOcc [35], OpenOccupancy
062 [34], CGFormer [41], and FoundationSSC [9] improved
063 view transformation, voxel attention, and semantic-
064 geometric fusion. These designs demonstrate the value
065 of explicit 3D representations.

066 OccuFly [14] enables a focused study of this trans-
067 fer problem. It provides aerial RGB images, metric
068 depth maps, camera calibration, poses, altitude meta-
069 data, and semantic occupancy grids across multiple
070 scenes and flight heights. Unlike 2D UAV segmen-
071 tation datasets such as UAVid [24] or reconstruction-
072 oriented resources such as UrbanScene3D [22], OccuFly
073 supports joint evaluation of monocular depth estima-

tion (MDE), multiview geometry, and SSC from real aerial observations. We use it to ask whether modern foundation representations provide transferable geometric and semantic priors for UAV 3D perception, and whether calibrated neighboring views improve SSC beyond monocular depth priors.

We investigate frozen V-JEPA 2.1 [25] image and video features for aerial MDE and depth-aware SSC, and add frozen DINOv3 [32] controls under the same protocol. V-JEPA-style models learn by predicting latent representations rather than reconstructing pixels [5, 6]; V-JEPA 2.1 further emphasizes dense predictive supervision, deep self-supervision, and unified image/video tokenization. DINOv3 is a particularly relevant control because it targets dense image transfer and includes aerial imagery in its evaluation. We freeze every backbone and vary only the decoder, metadata conditioning, temporal adapter, or lifting mechanism. Claims specific to V-JEPA are made only where the matched DINOv3 control differs; shared behavior is reported as a property of large frozen self-supervised ViTs.

Our evaluation has three parts. First, we compare linear probes, DPT-style multiscale decoders [30], Simple Feature Pyramid (SFP) adapters inspired by ViTDet [19], and Feature-wise Linear Modulation (FiLM) [27] using altitude and camera intrinsics. This separates linear depth accessibility, dense reconstruction capacity, final-layer pyramid adaptation, and metric scale conditioning. Second, we compare RGB features with spatio-temporal V-JEPA 2.1 video features using a dense temporal-attention adapter that converts tubelet tokens into target-frame maps. Third, we evaluate SSC variants inspired by CGFormer and FoundationSSC, replacing or augmenting their geometry pathways with V-JEPA-derived depth probabilities and a pose-aware multiview stereo (MVS) cost-volume branch based on calibrated UAV views.

Figure 1 gives the complete data flow, from RGB/video frames and calibration through frozen tokens, depth decoding, lifting, and semantic completion. Our contributions are:

- A matched frozen-backbone study of V-JEPA 2.1 and DINOv3 for aerial metric depth, separating backbone effects from decoder effects.
- A controlled comparison of last-layer linear probes, four-level DPT, final-layer SFP, and altitude/intrinsics-conditioned FiLM-DPT; an anomalous historical DPT result is replaced by a reproducible rerun.
- An analysis of spatio-temporal feature transfer through dense temporal attention over V-JEPA 2.1 video tokens.

- A depth-to-SSC study that injects V-JEPA-derived depth maps and depth distributions into CGFormer- and FoundationSSC-inspired aerial SSC variants.
- A diagnostic SSC study, including per-class results, that tests monocular and calibrated multiview geometry without claiming a new general SSC architecture.

2. Related Work

2.1. Monocular depth estimation and dense prediction decoders

Monocular depth estimation is ambiguous because metric depth must be inferred from perspective, texture, object scale, layout, and learned priors. Early deep methods used coarse-to-fine CNNs and scale-aware objectives [10]. Later work improved transfer through stronger encoders, multiscale dense decoders, and broader supervision. MiDaS showed the benefit of mixing heterogeneous datasets [29]; DPT reassembled transformer tokens into multiscale image-like features for dense prediction [30]; ZoeDepth combined relative-depth generalization with metric specialization [7]; and Depth Anything/Depth Anything V2 scaled monocular depth training with large pseudo-labeled and synthetic data [37, 38]. Aerial depth remains a distinct transfer problem because altitude changes scale, many images lack ground-level horizon cues, repetitive roofs and vegetation reduce texture, and far-range depth errors can degrade voxel lifting. We therefore compare frozen linear probes, DPT-style decoders, SFP adaptation inspired by ViTDet [19], and altitude/intrinsics FiLM conditioning [27] on the same V-JEPA 2.1 features.

2.2. Multiview stereo and cost-volume geometry

Multiview stereo (MVS) estimates depth by matching calibrated observations across views. MVSNet constructs plane-sweep cost volumes, regularizes them with 3D convolutions, and infers depth from a distribution over depth hypotheses [39]. Recurrent MVSNet reduces memory by processing hypotheses recurrently [40]. This projective matching principle is directly relevant to UAV SSC: when poses are available, target/source correspondence can provide geometric evidence complementary to monocular priors. Our Pose-MVS variants adapt this idea to frozen V-JEPA 2.1 video features by predicting a categorical depth distribution aligned with FoundationSSC lifting bins, then using that distribution to place semantic context into the OccuFly voxel grid.

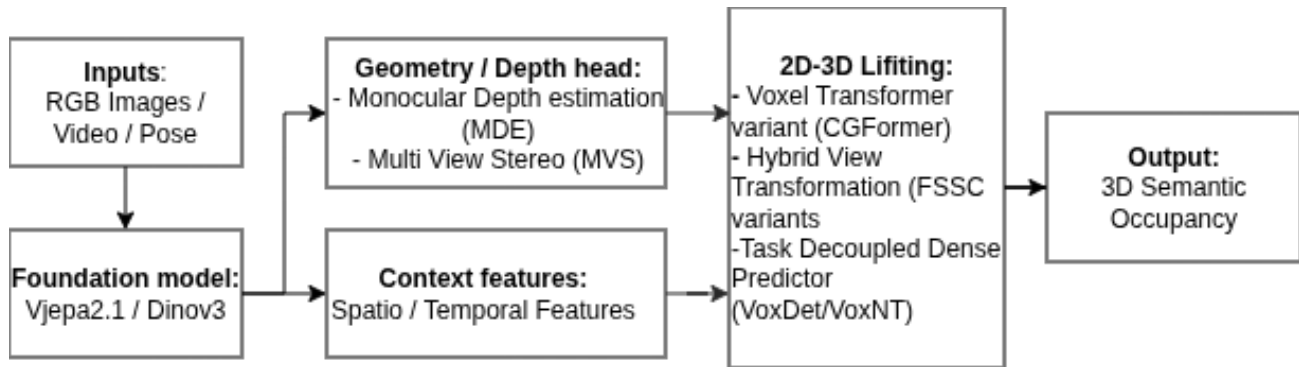


Figure 1. Generalized end-to-end architecture of the controlled experiment space. RGB images or video clips and camera metadata are processed by frozen visual foundation backbones, task-specific depth and geometry modules, depth-aware voxel lifting, and semantic scene-completion heads to produce metric depth, occupancy, and semantic voxel predictions.

2.3. Self-supervised visual foundation models

Self-supervised visual foundation models learn transferable representations without task-specific labels. Masked autoencoders scale ViTs through high-ratio masked image reconstruction [15]. I-JEPA predicts target-block representations from context representations in latent space [4]. DINOv2 produces general-purpose visual features from large-scale self-supervision [26], and DINOv3 emphasizes high-resolution dense features and transfer across natural and aerial imagery [32]. Video representation learning extends these ideas to temporal structure: V-JEPA predicts masked spatio-temporal features in latent space [6], V-JEPA 2 scales the approach to large image/video data [5], and V-JEPA 2.1 modifies the recipe toward dense image/video features [25]. Probe3D and Lexicon3D establish frozen multi-backbone protocols for testing 3D information in visual representations [11, 36]. We follow this principle with matched V-JEPA 2.1 and DINOv3 heads and reserve backbone-specific claims for differences that survive this control.

2.4. Camera conditioning and aerial benchmarks

Camera-aware depth decoding predates our FiLM design: CAM-Convs inject calibrated camera parameters into depth networks [12], while Metric3D canonicalizes camera geometry to improve metric transfer [16]. Our narrower question is whether altitude and intrinsics help decode metric depth from a frozen representation when decoder capacity is fixed. Aerial perception is also not limited to OccuFly. MidAir, WildUAV, UseGeo, AerialMetric, RuralScapes, DronesCapes, and MESSI cover complementary synthetic and real flights, geolocalization, metric geometry, rural scenes, and elevation changes [1–3, 13, 31, 42]. OccuFly is used because it jointly supports our standard-split MDE and

voxel-SSC evaluation; cross-dataset real-flight transfer remains necessary future work.

2.5. 2D-to-3D lifting and camera-based semantic scene completion

Semantic scene completion jointly predicts free/occupied voxels and occupied semantic labels. SSCNet introduced the task from single-view depth [33], and MonoScene lifted monocular RGB features into 3D [8]. Outdoor camera-based SSC then moved toward sparse voxel queries, deformable attention, and explicit view transformation. VoxFormer initializes visible voxel queries using estimated depth [21]; TPVFormer represents 3D space with three orthogonal planes [17]; and SurroundOcc/OpenOccupancy extend occupancy prediction to surround-view driving settings [34, 35].

Depth-aware lifting is a recurring theme. Lift-Splat-Shoot predicts a ray-wise depth distribution and splats image features into a shared spatial grid [28]; BEVDepth shows that explicit depth supervision improves camera-based BEV perception [20]. CGFormer addresses depth ambiguity with context-dependent voxel queries and 3D pixel-space deformable cross-attention [41]. FoundationSSC decouples semantic and geometric priors at both source and pathway levels before hybrid 3D fusion [9]. Our SSC experiments follow this line but focus on aerial observations: V-JEPA 2.1 features provide semantic context, while monocular depth, video aggregation, and pose-aware MVS provide alternative geometry priors under a shared OccuFly protocol.

3. Method

We separate four questions: whether geometry is accessible from a frozen representation (RQ1), whether

metadata helps beyond decoder capacity (RQ2), whether calibrated motion compensation improves temporal geometry (RQ3), and whether SSC is limited by depth, lifting, or semantic taxonomy (RQ4). Each OccuFly sample provides RGB I_t , metric depth D_t , valid mask M_t , occupancy labels Y_t , intrinsics K_t , altitude h_t , and world-to-camera pose T_t . Unless stated otherwise, backbones are frozen and only the named head is trained.

3.1. Frozen features and monocular depth

V-JEPA 2.1 ViT-G receives 384×384 images and exposes 24×24 , 1664-channel maps from blocks $\{11, 23, 37, 47\}$. The linear probe uses only block 47 and a batch-normalized 1×1 head. Four-level DPT reassembles and fuses all selected blocks; SFP-DPT instead creates a pyramid from the final block. Matched DINOv3 controls use the same split, resolution, training budget, objective, and relative transformer depths.

All heads predict $K_D = 256$ metric-depth logits z and the expectation

$$p_{k,u,v} = \text{softmax}_k z_{k,u,v}, \quad \hat{D}_{u,v} = \sum_{k=1}^{K_D} p_{k,u,v} d_k. \quad (1)$$

FiLM tests camera conditioning without changing the backbone. A small MLP embeds

$$m_t = [(h_t - 40)/10, f_x/W, f_y/H, c_x/W - 0.5, c_y/H - 0.5], \quad (2)$$

and modulates each DPT level R_l as $\tilde{R}_l = R_l \odot (1 + \gamma_l(m_t)) + \beta_l(m_t)$. Zero-initialized affine projections begin at the unconditioned decoder. Thus DPT versus FiLM-DPT is the relevant capacity-matched comparison; the anomalous historical DPT run is excluded.

3.2. Temporal aggregation and pose-aware MVS

A four-frame target-centered neighborhood is encoded as video tokens. Dense Temporal Attention (DTA) learns one target map per spatial cell from the two temporal tubelets, with a residual initialized near temporal mean pooling. Because frame-index neighbors move with the UAV and timestamps are unavailable, DTA alone does not establish correspondence. We therefore treat it as an unaligned temporal baseline.

Pose-MVS retains separate target/source features and explicitly compensates camera motion. For target pixel $\mathbf{u}$ and candidate depth d_k , a 3D point is projected into source view i by

$$\mathbf{q}_{i,k} = \pi(K_i T_{t \rightarrow i} [d_k K_t^{-1} \mathbf{u}; 1]). \quad (3)$$

We sample the source feature at $\mathbf{q}_{i,k}$, correlate it with the target feature, aggregate two source views, and regularize the resulting plane-sweep volume with a small

3D CNN. Softmax over the same 256 bins yields both direct metric depth and the probability distribution used for lifting. Pose-MVS-FiLM additionally conditions the output logits on m_t . Direct MDE metrics distinguish a weak matcher from a failure in later SSC fusion.

3.3. Depth-to-SSC and VoxNT

SSC separates semantic context from geometry. A monocular or Pose-MVS depth distribution lifts 2D context into the $192 \times 128 \times 128$ OccuFly grid; FoundationSSC/CGFormer-style 3D fusion then predicts empty plus semantic classes. Oracle-depth lifting bounds the loss attributable to depth.

VoxDet variants add a spatial encoder, three-dimensional feature pyramid, and VoxNT supervision. VoxNT predicts six directional offsets toward nearby semantic structure, providing a local geometric target before final voxel classification. Our variants isolate context source, volume width, corrected full-resolution VoxNT targets, long-tail semantic losses, and scheduled depth gradients. The strongest completed configuration uses frozen V-JEPA 2.1 RGB features as semantic context and a pretrained FiLM-DPT depth head to produce the dense depth and depth-bin probabilities used for lifting. The lifted 128-channel volume is decoded with full-resolution VoxNT supervision on localizable classes, class-balanced focal/dice semantic losses, an auxiliary occupancy/semantic head, a teacher-depth loss, and a cosine learning-rate schedule. Gradients from SSC back into the depth pathway are frozen for one epoch and then ramped over three epochs, which lets the voxel decoder stabilize before weakly adapting the depth distribution. A vehicle-instance extension keeps this SSC objective unchanged but initializes from the best VoxDet checkpoint and adds a low-resolution vehicle-only head: a three-channel center-offset branch teaches vehicle voxels to vote toward pseudo-instance centers, and an eight-dimensional discriminative embedding branch pulls voxels from the same vehicle together while pushing distinct pseudo-instances apart. Instance supervision is applied only on frames with at least 400 ground-truth vehicle voxels; other frames still train the semantic SSC objective. UAV11 retains empty and merges the 21 occupied labels into ten UAV-oriented groups; it tests how much fine-class sparsity and semantic overlap depress macro IoU, but is not directly comparable to the original taxonomy. Full architectural equations, tensor contracts, taxonomy maps, and per-class supports are supplied separately.

4. Experimental Setup and Results

4.1. Dataset and Tasks

We evaluate monocular depth estimation (MDE) and semantic scene completion (SSC) on OccuFly [14]. The standard split uses scenes 01–05 for training, scenes 06–07 for validation, and scenes 08–09 for testing across altitudes of 30, 40, and 50 m. The depth split contains 14,804 training frames, 1,965 validation frames, and 3,842 test frames. RGB inputs are resized to 384×384 ; metric depth is clipped to $[0.001, 80]$ m and evaluated only over finite positive pixels that remain valid after mask-aware resizing. Video variants use consecutive frame-index neighborhoods because the released pre-process files provide frame indices and poses but no measured capture FPS.

For SSC, OccuFly labels are defined on a $192 \times 128 \times 128$ voxel grid in WHD order. The grid spans $x \in [-48, 48]$ m, $y \in [-32, 32]$ m, and $z \in [0, 64]$ m, with 0.5 m voxels. Unless stated otherwise, experiments are run on an NVIDIA DGX node with 8 NVIDIA H100 80GB GPUs.

Controlled protocol and cost. Backbones are frozen; training updates only the named head. The matched V-JEPA/DINO comparisons use the same input resolution, official split, seed 0, per-GPU batch size 1 (global batch 8), 20-epoch budget, loss, and checkpoint criterion. Feature levels are matched by relative transformer depth. Exact decoder equality is impossible because V-JEPA 2.1 and DINOv3 expose 1664- and 1280-channel tokens: the corresponding linear heads contain 429,568 and 330,496 trainable parameters. DINOv3 ViT-H+/16 contains 840.6M frozen parameters. We report this deviation rather than projecting one backbone through an extra unmatched adapter. All current results are single-seed; margins are treated as descriptive until matched seeds are completed. Eight H100s accelerate the frozen-feature protocol but are not an onboard deployment claim.

4.2. MDE Benchmarking Results

Table 1 compares depth models on the standard OccuFly test split. We distinguish final test metrics from validation RMSE throughout. The historical plain-DPT row (11.471 m) was anomalous and its raw checkpoint is no longer auditable; we therefore replace it with the reproducible matched rerun (6.730 m). This correction reverses the earlier, implausible ordering: four-level DPT now improves over the last-layer linear probe. Under the matched FiLM-DPT protocol, DINOv3 reaches 4.358 m RMSE versus 4.770 m for V-JEPA 2.1, an 8.6% reduction. DINOv3 also im-

Table 1. OccuFly test MDE. In-house single-frame rows use frozen backbones at 384^2 resolution and the official 3,842-frame scene split. DINOv3 rows match the V-JEPA training protocol for the corresponding head type. ‡ denotes the 3,830-frame context-valid subset used by the video MVS evaluator. Published DepthAnything2-OccuFly values follow OccuFly [14]. Dashes denote planned but unfinished runs. Best values are bolded per metric among completed rows.

Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
V-JEPA 2.1 linear probe	0.651	0.959	0.997	0.188	7.795	6.447	0.225
V-JEPA 2.1 four-level DPT	0.747	0.970	0.997	0.162	6.730	5.411	0.200
V-JEPA 2.1 FiLM-DPT	0.816	0.970	0.998	0.131	4.770	3.457	0.166
DINOv3 linear probe	0.782	0.980	0.998	0.156	6.097	5.066	0.186
DINOv3 FiLM-DPT	0.848	0.968	0.998	0.121	4.358	3.209	0.161
V-JEPA 2.1 DTA-MVS (pending)	—	—	—	—	—	—	—
V-JEPA 2.1 DTA-FiLM-MVS†	0.768	0.930	0.987	0.175	6.406	4.811	0.211
DepthAnything2-OccuFly [14]	0.834	0.976	0.999	0.134	4.844	4.193	0.112

proves AbsRel, MAE, SILog, and δ_1 . The DINOv3 linear probe reaches 6.097 m RMSE, between the V-JEPA linear probe and the V-JEPA four-level DPT head, confirming that decoder capacity remains an important factor even with stronger frozen features. The pose-aware video checkpoint evaluated here is DTA-FiLM-MVS, not plain DTA-MVS: its multiview cost-volume output is FiLM-conditioned by altitude and intrinsics. It is evaluated on the 3,830 context-valid test frames, excluding the first and last frame of each scene–altitude sequence; the single-frame rows use all 3,842 test frames. We therefore report it as a diagnostic row and leave the unconditioned DTA-MVS ablation pending. Overall, these results show that the decoder and camera conditioning transfer across large frozen ViTs and that the strongest MDE result is not specific to V-JEPA 2.1.

The previous altitude-interpolation table contained one method and reused scenes across altitude partitions, so it could not isolate conditioning from scene memorization. We retain that result only as an explicitly exploratory appendix diagnostic (Appendix Table 7); it is not used to support the FiLM claim. A valid causal comparison requires matched FiLM/no-FiLM heads and scene-disjoint training and testing.

Appendix Table 3 shows that the video FiLM-DPT branch is strongest at 30 m, while the RGB FiLM-DPT branch is stronger at 40 m and 50 m. The corresponding scene-wise breakdown is provided in Appendix Table 4: video helps on scene_08, but RGB FiLM-DPT is stronger on scene_09.

4.3. SSC Benchmarking Results

RQ4: Is low SSC accuracy primarily a geometry or taxonomy problem? OccuFly is extremely imbalanced: empty voxels dominate the grid, building is the largest occupied class, and several safety-relevant classes have

Table 2. OccuFly test SSC comparison. † denotes results retained from the earlier ACCV26 evaluator. The canonical VoxDet rows use the same 3,842-frame test re-evaluation, one global confusion matrix, GT-present non-empty-class mIoU, and ignore label 255 excluded. ‡ marks a canonical test evaluation of a validation-selected checkpoint from a training job that stopped before its configured 20 epochs. Architecture names expose the changed context, geometry, and optimization components.

Method	SC IoU †	SSC mIoU †
Published OccuFly baselines		
Symphonies [18]	13.68	0.58
DISC [23]	29.52	2.04
Earlier ACCV26 context-geometry variants (†)		
CGFormer + DAV2 depth prior [†]	36.77	3.05
CGFormer + FiLM-DPT depth prior [†]	36.51	3.31
FSSC: RGB-DPT context + FiLM-DPT depth [†]	37.68	3.46
FSSC: FiLM-DPT context + FiLM-DPT depth [†]	35.47	1.73
FSSC: video DTA-DPT context + DTA-MVS depth [†]	24.55	0.17
FSSC: video DTA-DPT context + DTA-FiLM-MVS depth [†]	32.55	2.36
Canonical VoxDet component ablations		
CGFormer + DAV2 depth prior	9.93	1.37
FSSC: RGB-DPT context + FiLM-DPT depth	34.42	3.48
VoxDet: ResNet context + FiLM-DPT depth	38.85	2.85
VoxDet: RGB-DPT context + FiLM-DPT depth	38.87	3.47
VoxDet V5-V2: RGB-DPT context + FiLM-DPT depth + full-res VoxNT + long-tail loss + scheduled depth gradients [†]	45.77	4.62
VoxDet V5-V2: RGB-DPT context + DAV2 geometry + full-res VoxNT + long-tail loss [†]	44.82	4.35
MVSF++/D3 + VoxDet V5-V2: RGB-DPT context + frozen MVS posterior + full-res VoxNT + long-tail loss [†] (3 seeds)	49.33 ± 1.17	5.22 ± 0.35

only negligible support. Consequently, aggregate voxel accuracy can remain high while macro semantic IoU is low. Similar-looking structures further fragment supervision across fine labels (e.g., road/walkway/parking surfaces, or building/roof/construction), so errors can reflect semantic ambiguity in addition to lifting error. We therefore report SC IoU, GT-present SSC mIoU, fixed-denominator mIoU, frequency-weighted IoU, and per-class support rather than interpreting one headline number alone.

Table 2 restores the complete earlier comparison alongside the newly verified VoxDet ablations. Superscript † identifies ACCV26-era results produced by the earlier evaluator; the canonical VoxDet block is re-evaluated on one 3,842-frame manifest, with ‡ flagging checkpoints from interrupted training jobs. We keep the protocols visibly separated rather than treating their absolute values as directly comparable. Pose-MVS is evaluated separately because its earlier checkpoint was unavailable. Because absolute SSC mIoU remains low and several margins are small, we interpret these experiments as diagnostics of geometry, lifting, and taxonomy rather than as a state-of-the-art claim.

VoxDet/VoxNT finding. VoxNT predicts six-direction offsets toward local semantic structure and supplies a geometry-aware auxiliary signal before the final 3D decoding. Combining the corrected VoxDet V5-V2 decoder, long-tail objective, and full-resolution VoxNT with frozen DINOv3 MVSFormer++ geometry gives the strongest canonical test completion result. Across seeds 0/1/2 it reaches 49.33 ± 1.17 SC IoU, 5.22 ± 0.35 union-present SSC mIoU, 5.60 ± 0.43 GT-present mIoU, and 4.80 ± 0.36 fixed-21 mIoU (mean $\pm$ sample SD); validation selects

epochs 13/12/11. Every seed exceeds the single historical MVS FSSC control in SC IoU and union-present mIoU, showing persistence of the observed direction without constituting a paired significance test. The native MVS provider remains frozen and is executed without gradients, so the scheduled SSC-to-depth gradient used by the FiLM-DPT V5-V2 recipe is not active in this treatment. The FiLM-DPT geometry configuration reaches 45.77/4.62, while the matched DAV2-geometry configuration reaches 44.82/4.35. The larger completion gain than semantic gain remains consistent with VoxNT and stronger multiview geometry improving occupied structure while rare-class recognition remains limited by supervision. Per-seed and class-wise results are provided in the appendix.

UAV11 remapping diagnostic. We additionally merge the 21 occupied labels into ten UAV-oriented semantic groups plus empty (UAV11): constructed structure, transport surface, natural ground, low vegetation, tree, ground/construction obstacle, vulnerable road user, vehicle, thin infrastructure, and water. With the full-res VoxNT, long-tail-loss, scheduled-depth-gradient architecture trained from scratch, the best validation SSC mIoU increases from 5.64 on the original fine taxonomy to 9.32 on UAV11; validation SC IoU is 29.42. This is not a like-for-like benchmark improvement because the label space changes. It instead demonstrates that part of the low fine-class mIoU comes from sparse support and visually overlapping subclasses. Rare merged groups can remain unresolved—the UAV11 vehicle group has only 53 validation voxels—so remapping reduces, but does not remove, the long-tail problem. Full mappings, supports, and per-class scores are provided in the supplement.

RQ3: Where do temporal features help? Across the current MDE and SSC geometry experiments, temporal aggregation has only a small and inconsistent effect. Video FiLM-DPT is strongest at 30 m and on scene 08, but RGB is stronger at 40/50 m and on scene 09. Uncompensated camera motion makes snapshot-token aggregation an ambiguous test of temporal representation quality; Pose-MVS therefore preserves view identity and explicitly warps source features with calibration. We do not conclude that temporal features are unhelpful in general. OccuFly provides short sampled neighborhoods rather than continuous video, so benefits may emerge for dense streams with stable temporal correspondence, motion history, or dynamic cues.

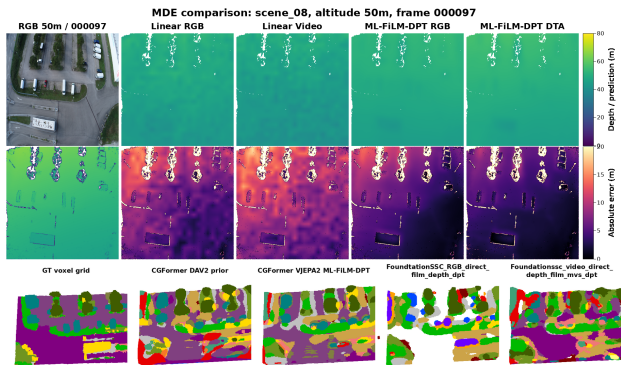


Figure 2. Qualitative OccuFly depth and SSC example for scene 08 at 50 m altitude, frame 000097.

5. Discussion and Conclusion

What is linearly accessible? The frozen V-JEPA 2.1 last-layer probe reaches 7.795 m RMSE and 0.225 SILog on the standard test split. The corrected four-level DPT improves these to 6.730 m and 0.200, respectively. Thus the data do not support the earlier conclusion that a linear probe outperforms dense decoding. Instead, aerial geometry is present but benefits from recovering and fusing intermediate spatial structure. The matched DINOv3 linear probe will determine whether this accessibility is specific to V-JEPA or shared by large frozen ViTs.

What does FiLM establish? The strongest traceable V-JEPA 2.1 FiLM-DPT reaches 4.770 m RMSE, 0.131 AbsRel, and 0.166 SILog. Relative to the corrected plain DPT, this is consistent with altitude/intrinsics helping metric scale recovery. However, the available FiLM result comes from a sweep rather than a fully matched multi-seed pair, so the current evidence supports association, not a precise causal effect size. We no longer compare FiLM against the failed 11.471 m historical DPT checkpoint. DepthAnything2 remains stronger in SILog and threshold accuracy even when V-JEPA FiLM-DPT has slightly lower RMSE, indicating a tradeoff between metric error and scale-invariant consistency.

Where do video and SSC help? Temporal aggregation is altitude- and scene-dependent and does not consistently beat RGB: it helps at 30 m and on scene 08, but loses at 40/50 m and on scene 09. Since OccuFly exposes short sampled neighborhoods rather than continuous video, this negative result does not rule out benefits from dense streams, motion history, or dynamic cues. Pose-aware warping is the appropriate control

for camera motion and is evaluated directly in depth space. For SSC, the FiLM-DPT VoxDet V5-V2 recipe with full-resolution VoxNT, long-tail loss, and scheduled depth gradients reaches 45.77/4.62 SC IoU/SSC mIoU. Applying its compatible decoder, objective, and full-resolution VoxNT components to frozen DINOv3 MVSFormer++ geometry reaches 49.33 ± 1.17 SC IoU and 5.22 ± 0.35 union-present mIoU across three seeds (5.60 ± 0.43 GT-present; 4.80 ± 0.36 fixed-21). Every seed exceeds the single historical MVS control in the directly comparable fields, while the unrepeated control prevents a paired significance claim. The MVS posterior remains frozen, so scheduled depth gradients are not active in this treatment. UAV11 remapping reaches 9.32 validation mIoU versus 5.64 for the original fine taxonomy, showing that sparse and visually overlapping labels materially depress macro IoU; the changed label space makes this a diagnostic rather than a direct benchmark gain.

Limitations and conclusion. This is a frozen-feature study on one primary joint MDE/SSC dataset. It does not establish onboard efficiency, end-to-end finetuning behavior, cross-dataset real-flight transfer, statistical significance, or direct Pose-MVS depth quality. The altitude interpolation diagnostic reuses scenes across elevations and is therefore not causal evidence for conditioning. Within these limits, the corrected results show that large frozen self-supervised ViTs contain useful aerial geometry; decoder structure and calibrated metadata determine how effectively it becomes metric depth, while downstream SSC gains depend more strongly on geometry-context alignment than on swapping one depth prior.

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A. Additional Depth Results

Table 3. OccuFly test MDE comparison grouped by altitude, with Metric3Dv2-OccuFly and DepthAnything2-OccuFly altitude-wise values appended from OccuFly Table 12 [14]. Light pink marks the best RMSE within each altitude group.

Group	Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
30	Linear Probe RGB	0.743	0.952	0.986	0.205	6.280	5.391	0.176
	Linear Probe Video 4 frame clip	0.512	0.933	0.979	0.266	7.942	7.002	0.177
	SFP_FiLM_DPT	0.889	0.961	0.991	0.111	4.022	2.847	0.166
	DPT	0.415	0.889	0.940	0.295	9.505	8.238	0.350
	FiLM_DPT_RGB	0.917	0.980	0.999	0.097	3.448	2.571	0.132
	DTA_FiLM_DPT_Video	0.944	0.982	0.999	0.081	2.939	2.051	0.124
	Metric3Dv2-OccuFly [14]	0.460	0.757	0.992	0.347	10.917	10.215	0.101
	DepthAnything2-OccuFly [14]	0.919	0.982	0.998	0.103	3.108	2.666	0.086
40	Linear Probe RGB	0.730	0.943	0.998	0.180	5.851	4.804	0.217
	Linear Probe Video 4 frame clip	0.720	0.924	0.996	0.185	5.763	4.510	0.207
	SFP_FiLM_DPT	0.729	0.911	0.988	0.186	5.665	4.056	0.220
	DPT	0.577	0.794	0.905	0.239	9.106	6.916	0.391
	FiLM_DPT_RGB	0.761	0.945	0.995	0.168	4.804	3.588	0.186
	DTA_FiLM_DPT_Video	0.737	0.938	0.997	0.173	5.199	3.880	0.195
	Metric3Dv2-OccuFly [14]	0.209	0.802	0.981	0.395	13.138	12.625	0.976
	DepthAnything2-OccuFly [14]	0.795	0.957	0.998	0.148	4.486	3.823	0.149
50	Linear Probe RGB	0.524	0.978	1.000	0.192	5.981	4.728	0.982
	Linear Probe Video 4 frame clip	0.741	0.994	1.000	0.161	8.244	7.158	0.167
	SFP_FiLM_DPT	0.858	0.995	1.000	0.110	5.038	3.777	0.182
	DPT	0.375	0.754	0.872	0.285	11.243	12.179	0.371
	FiLM_DPT_RGB	0.873	0.995	1.000	0.113	4.872	3.879	0.143
	DTA_FiLM_DPT_Video	0.817	0.998	1.000	0.122	5.213	4.165	0.176
	Metric3Dv2-OccuFly [14]	0.283	0.831	0.987	0.379	15.373	14.950	0.734
	DepthAnything2-OccuFly [14]	0.844	0.997	1.000	0.129	5.985	5.261	0.114

Table 4. OccuFly test MDE comparison grouped by scene. Light pink marks the best RMSE within each scene group.

Group	Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
scene 08	Linear Probe RGB	0.985	0.999	0.999
	Linear Probe Video 4 frame clip	0.983	0.999	0.999
	SFP_FiLM_DPT	0.998	1.000	1.000
	DPT	0.976	0.996	0.997
	FiLM_DPT_RGB	0.996	0.999	0.999
	DTA_FiLM_DPT_Video	0.997	0.997	0.997
	Metric3Dv2-OccuFly [14]	0.996	0.996	0.996
	DepthAnything2-OccuFly [14]	0.997	0.997	0.997
scene 09	Linear Probe RGB	0.947	0.999	0.999
	Linear Probe Video 4 frame clip	0.941	0.999	0.999
	SFP_FiLM_DPT	0.935	0.999	0.999
	DPT	0.734	0.866	0.866
	FiLM_DPT_RGB	0.960	0.999	0.999
	DTA_FiLM_DPT_Video	0.958	0.999	0.999
	Metric3Dv2-OccuFly [14]	0.958	0.999	0.999
	DepthAnything2-OccuFly [14]	0.958	0.999	0.999

B. V07b Repeated-Seed Results

Seeds 0, 1, and 2 use the same frozen MVS checkpoint, data split, training contract, and validation-only selection rule. Table 5 reports percentages and sample standard deviation ($n - 1$, $n = 3$). Fixed-21 mIoU is prediction-independent; union-present uses 19/19/20 classes across the three seeds and is retained for the historical control comparison.

C. Exploratory Altitude Interpolation

Table 7 records the historical altitude-interpolation diagnostic. Training uses 30 and 50 m samples, while evaluation uses 40 m samples. Because scenes recur across the protocol and only one conditioned method was evaluated, this is not a causal FiLM ablation and is not used to support the main conditioning claim.

D. Detailed SSC Results

Additional class-wise and altitude-wise SSC results are provided here to keep the main paper focused on the principal trends.

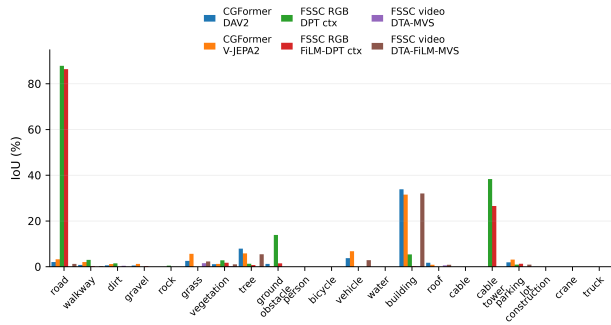


Figure 3. OccuFly SSC per-class IoU bar chart for the evaluated CGFormer and FoundationSSC variants.

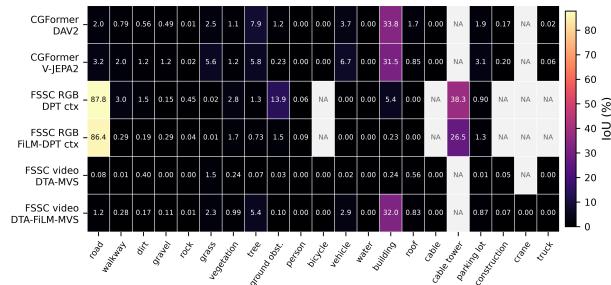


Figure 4. OccuFly SSC per-class IoU heatmap for the evaluated CGFormer and FoundationSSC variants.

Table 5. V07b test performance across seeds (%).

Seed	Epoch	SC IoU	GT-present mIoU	Fixed-21 mIoU	Union mIoU	SC precision	SC recall
0	13	48.0635	5.1116	4.3814	4.8426	62.6235	67.3974
1	12	49.5434	5.8380	5.0040	5.5308	63.6363	69.1083
2	11	50.3749	5.8589	5.0219	5.2730	62.6274	72.0269
Mean $\pm$ SD	–	49.3273 $\pm$ 1.1708	5.6028 $\pm$ 0.4256	4.8024 $\pm$ 0.3648	5.2154 $\pm$ 0.3477	62.9624 $\pm$ 0.5836	69.5109 $\pm$ 2.3409

Table 6. V07b occupied-class IoU across seeds (%). GT-absent classes are not assigned a semantic mean.

Class	Test GT	Seed 0	Seed 1	Seed 2	Mean $\pm$ SD
Road	Present	4.1026	5.8616	6.0519	5.3387 $\pm$ 1.0747
Walkway	Present	3.2933	4.5309	4.6426	4.1556 $\pm$ 0.7489
Dirt	Present	0.9252	1.8369	1.1948	1.3190 $\pm$ 0.4684
Gravel	Present	1.6976	2.9167	3.3397	2.6513 $\pm$ 0.8526
Rock	Present	0.0103	0.0190	0.0125	0.0140 $\pm$ 0.0045
Grass	Present	5.7403	6.9937	7.1410	6.6250 $\pm$ 0.7697
Vegetation	Present	2.6840	3.5136	3.1042	3.1006 $\pm$ 0.4148
Tree	Present	10.5557	10.6668	12.0986	11.1071 $\pm$ 0.8605
Ground obstacle	Present	1.4844	2.0339	2.0121	1.8435 $\pm$ 0.3112
Person	Present	0	0	0	0 $\pm$ 0
Bicycle	Present	0	0	0	0 $\pm$ 0
Vehicle	Present	8.3794	9.3942	9.4271	9.0669 $\pm$ 0.5956
Water	Absent	–	–	–	–
Building	Present	44.2941	47.2395	46.8691	46.1342 $\pm$ 1.6043
Roof	Present	1.9902	2.1467	1.8644	2.0004 $\pm$ 0.1414
Cable	Present	0	0	0	0 $\pm$ 0
Cable tower	Absent	–	–	–	–
Parking lot	Present	6.5386	7.4779	7.2042	7.0736 $\pm$ 0.4831
Construction	Present	0.2490	0.4162	0.3141	0.3264 $\pm$ 0.0843
Crane	Absent	–	–	–	–
Truck	Present	0.0641	0.0366	0.1842	0.0949 $\pm$ 0.0785

Table 7. Exploratory altitude-interpolation result. The protocol is not scene-disjoint and contains no matched unconditioned control.

Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
V-JEPA 2.1 RGB FiLM-DPT	0.9653	0.9997	0.9999	0.1021	4.7071	3.9570	0.1196

Table 8. OccuFly SSC benchmark comparison by altitude. Published Symphonies and DISC values follow OccuFly Table 4 [14]; CGFormer and FoundationSSC values are from the local SSC benchmark report. For our variants, light beige marks the best SC IoU and light pink marks the best SSC mIoU within each altitude group and the overall All group.

Altitude	Method	SC IoU $\uparrow$	SSC mIoU $\uparrow$
Published OccuFly Table 4 baselines			
50 m	Symphonies [18]	15.88	0.58
50 m	DISC [23]	31.10	2.20
40 m	Symphonies [18]	10.71	0.52
40 m	DISC [23]	27.85	1.77
30 m	Symphonies [18]	13.22	0.76
30 m	DISC [23]	26.88	2.23
All	Symphonies [18]	13.68	0.58
All	DISC [23]	29.52	2.04
Our CGFormer and FoundationSSC variants			
30 m	CGFormer DAV2 prior	36.10	4.22
30 m	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	36.70	4.39
30 m	FSSC RGB ctx DPT, depth FiLM-DPT	33.21	5.16
30 m	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	6.74	0.26
30 m	FSSC video ctx DTA-DPT, depth DTA-MVS	2.84	0.18
30 m	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	35.59	5.06
40 m	CGFormer DAV2 prior	38.42	2.86
40 m	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	29.05	2.12
40 m	FSSC RGB ctx DPT, depth FiLM-DPT	27.85	4.08
40 m	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	22.15	1.69
40 m	FSSC video ctx DTA-DPT, depth DTA-MVS	12.30	0.87
40 m	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	29.22	3.97
50 m	CGFormer DAV2 prior	35.56	2.81
50 m	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	42.78	4.06
50 m	FSSC RGB ctx DPT, depth FiLM-DPT	21.16	2.53
50 m	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	19.51	0.75
50 m	FSSC video ctx DTA-DPT, depth DTA-MVS	5.21	0.21
50 m	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	22.28	2.59
All	CGFormer DAV2 prior	36.77	3.05
All	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	36.51	3.31
All	FSSC RGB ctx DPT, depth FiLM-DPT	37.68	3.46
All	FSSC RGB ctx FiLM-DPT, depth FiLM-DPT	35.47	1.73
All	FSSC video ctx DTA-DPT, depth DTA-MVS	24.55	0.17
All	FSSC video ctx DTA-DPT, depth DTA-FiLM-MVS	32.55	2.36

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